Course Code	21IPC501JR	Course Name	Research Methodology	Course Category	C	Professional Core	L	T	P	C
							2	1	2	4

Pre-requisite Courses	Nil	Co-requisite Courses	Nil	Progressive Courses	Nil
Course Offering Department	College of Engineering and Technology	**Data Book / Codes/Standards**	Statistical data book and tables		

Course Learning Rationale (CLR):	The purpose of learning this course is to:
CLR-1:	Understand the different methodologies involved in research studies
CLR-2:	Familiarize with various types of academic writing and presentations
CLR-3:	Understand the significance of ethics in academic publications
CLR-4:	Familiarize with different design of experiments and optimization techniques
CLR-5:	Familiarize with approaches of analysing data

Course Outcomes (CO):	At the end of this course, learners will be able to:	Programme Outcomes (PO)		
		1	**2**	**3**
CO-1:	Acquire knowledge on basic research methodology	3	3	-
CO-2:	Write and present research contents	3	3	-
CO-3:	Recognise the values and ethic involved in scientific research	3	3	-
CO-4:	Apply different approaches for solving design of experiments and optimization problems	3	2	-
CO-5:	Analyse the applications of statistical and machine learning tools for data analysis	3	2	-

Module-1 – RESEARCH PREPARATION, PLANNING AND RESOURCES	16 Hours
Research – objectives, definitions – knowledge economy; Perspectives of stakeholders; Types of research; Critical & creative thinking; Generating research topic; Identifying broad research question and objectives – design of questionnaire; Characteristics of a good research – research design; Methods of scientific enquiry – theoretical, experimental and empirical; Hypothesis – formulation & testing. Sources of information – ICT enabled tools; Databases, repositories, public and private sources, indexes; Literature search – keyword; Backward and forward search; Quality measurement tools: journal metrics, author-level metrics; Meta-analysis of research findings; Literature review: grouping, analysing & comparison: Identification of research Methods; Intellectual Property Rights; Copyright – patent –geographical indication – industrial designs – technology transfer. PRACTICE: Identifying research topic and formulation of Research Statement; Conduction of literature review.	
Module-2 – ACADEMIC WRITING AND PRESENTATIONS	**14 Hours**
Preparation of project proposal for funding – Identification of funding agencies; Project proposals structure; Research report writing; Data visualization; Communication model – audience analysis; Identification of suitable journal; Conference presentation types: Oral, poster – difference in audience interaction – guideline for effective presentation; Synopsis – thesis – extended abstract – graphical and video abstract – short communications; Referencing – style – tools for referencing; Thesis writing: structure – preliminary pages, main body, references; Evaluation of Thesis – examiner reports – Oral defence. PRACTICE: Writing abstract and conclusion; Referencing in different styles.	
Module-3 – PUBLICATION ETHICS AND MISCONDUCTS	**12 Hours**

Ethics: definition; Ethics with respect to science and research - intellectual honesty and research integrity; Scientific misconducts: Falsification, Fabrication and Plagiarism (FFP); Redundant publications: duplicate and overlapping publications, salami slicing; Selective reporting and misrepresentation of data; Publication ethics: introduction and importance; Standards setting initiatives and guidelines: COPE, WAME; Conflicts of interest. Publication misconduct: definition, concept, unethical behaviour; Violation of publication ethics, authorship and contributor ship; Identification, complaints and appeals; Predatory publisher and journals. PRACTICE: Plagiarism checking and rectification.
Module-4 – DESIGN OF EXPERIMENTS AND OPTIMIZATION METHODS **16 Hours**
Design of experiments; Basic experimental designs; Completely randomized design; Randomized block design; Latin square design; Full factorial design - 2^2, 2^3 and 2^k; Fractional factorial designs; Taguchi methodology – Orthogonal array; S/N ratio – Taguchi method for optimization; Response surface method – multi response optimization. Global optimization techniques – evolutionary and swarm algorithms – examples and case studies. PRACTICE: Design of experiment – Taguchi method – Response surface method.
Module-5 – DATA ANALYSIS **17 Hours**
Basic probability distributions – Normal distribution – Binominal and Poisson distributions – Weibull and exponential distributions; Sampling: types - size, Sampling tests: T-test – F-test – $\chi 2$ test; Modelling – validation and verification; Simulation – techniques; ANOVA; Correlation analyses: Pearson and Spearman Correlations; Regression, classification and cluster analyses; Factor analysis; Discriminant analysis; Uncertainty and error analyses; Machine learning: supervised and unsupervised – examples and case studies. PRACTICE: Distribution; Sampling; Machine learning tools.

Learning Resources	***1. Ganesan, R. (2020), Research Methodology for Engineers, MJP Publishers, Chennai.*** ***2. Kothari, C.R. & Garg, G. (2022), Research Methodology, New Age International Publishers, New Delhi.*** ***3. Singh, U.P., Ahlawat, S. & Sharma, S. (2023), Research and Publication Ethics, S.Chand & Co. Ltd., New Delhi.*** *4. Mukerjee, S.P. (2020), A Guide to Research Methodology: An Overview of Research Problems, Tasks and Methods, CRC Press, New York.* *5. Bairagi, V. & Munot. M.V. (2019), Research Methodology: A practical and scientific approach, CRC Press, New York.* *6. Locharoenrat, K. (2017), Research Methodology for beginners, CRC Press, New York.* *7. Panneerselvam, R. (2014) Research Methodology, R., PHI Learning Pvt Ltd, New Delhi.* *8. Upagade, V. & Shende, A. (2010), A. Research Methodology, Vijay, S.Chand & Co. Ltd., New Delhi.* *9. Alvi, M.H. (2016), A Manual for Referencing styles in Research, University of Karachi, Pakistan.*	*10. Steneck, N.H. (2007). Introduction to the Responsible Conduct of Research. Office of Research Integrity. Available at: https://ori.hhs.gov/sites/default/files/rcrintro.pdf* *11. Adams, K.A., and Lawrence, E.K. (2019), Research Methods, Statistics and applications, Sage Publication, Thousand ORKS, California, USA* *12. Gastel, B., and Day, R.A. (2016), How to Write and Publish a Scientific paper, 8th Edition, Greenwood, California* *13. Kornuta, H.M., and Germaine, R.W., (2006), A concise guide to writing a Thesis or Dissertation, 2nd Edition, Taylor & Francis Group, New York* *14. Montgomery, D.C., (2013), Design and Analysis of Experiments, 8th edition, Wiley, New Delhi.* *15.. Box, G.E.P. and Draper N.R, (2007) Empirical Model-Building and Response Surfaces, John Wiley & Sons.* *16.. Antony, J., (2014), Design of Experiments for Engineers and Scientists, Second Edition, Elsevier, Amsterdam.* *17. Das, M.N. and Giri, N.C., (2003) Design and Analysis of Experiments, New Age International (P) Limited, New Delhi.* *18. Chopra, R. (2018), Machine Learning, Khanna Book Publishing, New Delhi.*

Learning Assessment

	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)				Summative Final Examination (40% weightage)	
		Formative CLA-1 Average of unit test (45%)		Life-Long Learning CLA-2 (15%)			
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	20%	-	-	10%-	20%	-
Level 2	Understand	20%	-	-	10%-	20%	-
Level 3	Apply	30%	-	-	40%-	30%	-
Level 4	Analyze	30%	-	-	40%-	30%	-
Level 5	Evaluate	-	-	-	-	-	-
Level 6	Create	-	-	-	-	-	-

	Total	*100 %*	*100 %*	*100 %*

Course Designers		
Experts from Industry	**Experts from Higher Technical Institutions**	**Internal Experts**
1.	1.	1.
2.	2.	2.